

Control-M SaaS Administration

Architecture, Agents, plug-ins and connection profiles, users and roles, authentication, API tokens, and system settings.

1. What changes in the SaaS model

In Control-M SaaS, BMC hosts and operates the control layer: the web interface, the database, and the SaaS Control-M/Server. Your team no longer installs, patches, backs up or recycles Control-M/EM or Control-M/Server. What stays on your side is everything that touches your systems: the Agents, the plug-ins on those Agents, the connection profiles that hold credentials, and who is allowed to do what.

Area	Self-hosted Control-M	Control-M SaaS
Control-M/EM and Control-M/Server	You install, patch, back up, recycle	Operated by BMC (24x7 SaaS operations)
Database	You manage (e.g. SQL Server, PostgreSQL, Oracle)	Managed by BMC; no DBO access
Upgrades of the control layer	Planned upgrade projects	Delivered continuously by BMC
Control-M/Agents and plug-ins	You install and upgrade	You install and upgrade (token-based)
Users and roles	You manage	You manage, usually via your IdP (SAML 2.0)
Interface	Control-M Web + Desktop client	Control-M Web at your tenant URL
Automation	Automation API	Automation API (token-authenticated)

The four web domains

- **Planning:** build and edit folders, jobs, calendars and resources in workspaces.
- **Monitoring:** viewpoints, services, alerts, and actions on running jobs.
- **Configuration:** Agents, Agent tokens, plug-ins, connection profiles, users, roles, IdP, API tokens, system settings.
- **Tools / Managed File Transfer:** site standards, user views, MFT search, and other utilities (depending on licensing).

Hybrid estates

Control-M SaaS also offers **Unified View**, which lets you connect self-hosted Control-M/Servers and Control-M for z/OS to the SaaS interface. This is mostly used during migration, so both estates can be watched from one place.

2. Agents

The Agent is the only Control-M component that runs on your hosts. It receives jobs from the SaaS Control-M/Server, runs them, and reports status and output back. Agent communication with SaaS is initiated from your side, so your network team needs to allow the Agent host to reach your tenant's endpoints. Check the current port and domain list in BMC's network requirements page before you install.

Agent tokens and tags

Every SaaS Agent installation needs an **Agent token**. The token connects the new Agent to your tenant and is tied to an **Agent tag**. A tag is a label that groups Agents under one authorization level. Each Agent has exactly one tag.

- A token can be reused for more installations with the same tag.
- A role that holds a tag pattern such as Fin* can create its own tag names at install time, e.g. FinDev or FinOps.
- Manage tokens under Configuration > Manage > **Agent Tokens**: generate, copy or delete them.
- Treat tokens like passwords. Store them in your vault, not in scripts or tickets.

Installing an Agent

1. Check prerequisites: supported OS, Java (a default JRE is bundled; you can use your own), and on Linux the pre-installation steps. For the Automation API CLI you need Python 3.8.4+ and pip 20.1.1+.
2. In Configuration, open **Agents** > Add Agent > **Install Agent** and pick the OS.
3. Windows: log on with administrative rights and download the installer. UNIX: copy the cURL or wget command and run it on the host.
4. Select an existing token or generate a new one (token name + tag), and copy it.
5. Set the Java environment variable, then run the installer from that same session.
6. Install interactively, silently with an XML file, or automatically through the Automation API (ctm provision saas::install).
7. Confirm the Agent appears in the Agents tab, then check installed-versions.txt in the Agent home for the right product code.

```
REM Silent install on Windows (parameters file generated from an interactive run)
<installer>.exe -silent C:\temp\agent_params.xml ^
    -BMC_INST_JAVA_HOME "C:\Program Files\Java\jre" -token <AGENT_TOKEN>

REM Install log
<installFolder>\BMCINSTALL\log\
```

The Agent name is filled in automatically: on Windows it combines hostname and Windows instance, on UNIX hostname and OS account. You can change it (alphanumeric plus - : _, starting and ending with an alphanumeric character).

Upgrading Agents

- Upgrade interactively, silently, or with the Automation API. API-driven upgrade needs Agent 9.0.20.280 or later and also upgrades the plug-ins on that Agent.
- Stop the Agent first if Control-M MFT or Control-M for SAP is installed on it, and start it afterwards.
- To avoid downtime, put several Agents with the same plug-ins in a **host group**. While one is stopped, jobs route to the others.

Everyday Agent administration

Task	Where / how
Check connectivity	Agents tab > select Agent > Ping
See Agent and plug-in versions	Agents tab columns; installed-versions.txt on the host
Group Agents for load balancing / failover	Host groups (jobs target the group instead of one host)
Remove a decommissioned server	Delete the Agent in the Agents tab, then uninstall on the host
Agent connected to self-hosted and SaaS servers	Multi-server Agent configuration (used in migrations)

3. Plug-ins and connection profiles

Plug-ins let an Agent run application-specific job types, such as file transfer (MFT), SAP, databases, cloud services (AWS, Azure, GCP), data platforms, and many others under Control-M Integrations. You deploy plug-ins onto Agents from the Configuration domain and can see which Agents have which plug-ins.

- **Connection profiles** hold the endpoint and credentials a plug-in job uses (for example an SFTP host, a database login, or a cloud account).
- **Centralized connection profiles** are defined once in the web interface and available to any Agent with the plug-in. Prefer these over local profiles, which live on one Agent.
- **Application Integrator** lets you build your own job type (for example around a REST API) without writing a full plug-in.
- Separate connection profiles per environment (dev / test / prod). Never point a non-production job at a production profile.

4. Users, roles and authentication

Authentication

You can create internal users, but BMC recommends connecting an **Identity Provider (IdP)** with SAML 2.0. Users then sign in with SSO at your tenant URL, and MFA is enforced by your IdP, not by Control-M.

1. Configuration > Identity Provider (IdP) > toggle on **Enable SAML 2.0**.
2. Exchange metadata with your IdP team (Audience URI / Service Provider Entity ID, sign-on URL, certificate).
3. Map IdP groups to Control-M roles, so access follows group membership.
4. Keep the **Emergency Users** list current. If the IdP is down, emergency users with Admin roles can log in through the Emergency URL. To edit that list you must temporarily disable the IdP configuration.

Tip

Internal users get a confirmation email with a temporary password. Use internal users sparingly, mainly for break-glass access, because they sit outside your joiner/mover/leaver process.

Roles and authorizations

Access is granted through roles. A user's rights are the combination of all their roles. Control-M ships three predefined roles; create your own for everything else. A role can restrict, among other things:

Authorization	What it controls
Control-M/Server	Which servers the role can run jobs on
Folders / jobs	Browse, update or full access, usually by folder or application name pattern
Run as users	Which OS accounts jobs may run under
Agents / Agent tags	Which Agents the role can see or install (tag patterns such as Fin*)
Monitoring actions	Hold, rerun, kill, set to OK, edit script/JCL, and so on
Resources, calendars, variables	Whether the role can view or change shared objects
Configuration	Access to Agents, plug-ins, connection profiles, users and roles

A practical role model

Role	Typical rights
CTM-Admin	Full Configuration access; small named group; members also on the Emergency Users list
CTM-Scheduler	Planning for assigned folders/applications; no Configuration access
CTM-Operator-L1	Monitoring: view, hold/release, rerun, confirm; no Set to OK, no script edit
CTM-Operator-L2	L1 rights plus kill, set to OK / Not OK, bypass, edit job in Monitoring
App-Support-<app>	View and limited actions only on their own application's folders
Automation (API token)	Only what the pipeline needs, e.g. deploy to dev folders

5. API tokens and Automation API

The Automation API is token-authenticated in SaaS. Create tokens under Configuration > **API Tokens** > Add Token, and attach only the roles the token needs. By default all of the creator's roles are selected, so narrow that list before saving.

```
# Register your SaaS environment in the ctm CLI (once per workstation)
ctm environment saas::add prod \
  https://<tenant>-aapi.<region>.controlm.com/automation-api \
  <API_TOKEN>
ctm environment set prod

# Quick checks
ctm config servers::get
ctm config server:agents::get <server>
```

Endpoint

Copy the exact endpoint from your tenant (shown when you create a token or in the API docs page of your tenant). The URL above only shows the shape.

6. System settings and governance

- **Site Standards:** rules that force naming and property conventions (e.g. job names must start with the application code). Attach them to folders so bad definitions are blocked at check-in.
- **User Views:** tailor what different user groups see in the interface.
- **Notification destinations:** shared mail lists and destinations that jobs reference by name.

- **Audit annotations:** require users to enter a reason when they take certain actions (for example Set to OK or Kill). Recommended for production.
- **Workload Policies:** cap how many jobs of a type run at once, or pause a set of jobs during maintenance.

Administrator checklist

Frequency	Check
Daily	All production Agents available; no Agents stuck Disconnected; failed plug-in deployments
Weekly	Unused or expired Agent and API tokens; new internal users; Agents on old versions
Monthly	Role membership review with application owners; Emergency Users list; connection profile passwords due to rotate
Per release	Read the Control-M SaaS release notes for UI or behaviour changes and brief the operators

7. Architecture in more detail

It helps to picture Control-M SaaS as three layers. Knowing which layer a problem sits in tells you straight away whether you fix it yourself or open a case with BMC.

Layer	Components	Who runs it	Typical issues
Presentation	Control-M Web at your tenant URL; Automation API endpoint	BMC	Login/SSO problems (often IdP-side), slow UI, API rate or token errors
Control	SaaS Control-M/Server, scheduling engine, database, New Day processing, statistics	BMC	Platform incidents, maintenance windows (announced by BMC)
Execution	Control-M/Agents, plug-ins, connection profiles, your hosts and applications	You	Agent down, host down, credentials expired, firewall/proxy changes, job logic errors

How a job travels

1. At New Day (or when ordered manually), the SaaS Control-M/Server places the job in the active environment.
2. When its time window, prerequisites and resources are satisfied, the server selects a host (or a host from the host group).
3. The job is sent to that Agent over the Agent's connection to the SaaS backend.
4. The Agent runs the command, script or plug-in call as the run-as user, and captures the output.
5. The Agent returns the completion status, exit code and output to the server.
6. The server performs post-processing: If-actions, events, notifications, output handling, statistics.

Why this matters

If step 3 or 5 cannot happen because the Agent is unreachable, the job sits in Wait Host (before submission) or eventually becomes Unknown/Disappeared (after submission). Almost every 'Control-M is broken' call at L1 is really an Agent or host problem in the execution layer.

What is stored where

Data	Location
Job and folder definitions, calendars, resources, roles	SaaS (BMC-managed database)
Active jobs, statuses, logs, statistics	SaaS
Job output (sysout)	Produced on the Agent; viewable through the web interface
Scripts called by jobs (unless embedded)	On your hosts; the job references their path
Plug-in binaries and local connection profiles	On the Agent
Centralized connection profiles (with credentials)	SaaS, encrypted
Agent install logs, Agent diagnostic logs	Agent home directory on the host

8. Agent host planning

Choosing where Agents go

- Install an Agent close to what it runs: on the application server itself, or on a utility server with network access to the target (for database, file transfer and cloud plug-ins).
- For each critical application, use at least two Agents in a **host group**, on different physical hosts or availability zones.
- Keep production and non-production Agents separate, with different Agent tags (for example PRD-* and NPD-*), so roles can be restricted by tag.
- Plan for patching: if a host reboots monthly, its jobs need either a second Agent in the host group or a maintenance window during which they are held.

Windows Agent considerations

Topic	Guidance
Install account	Local administrator rights are needed to install the Agent and its Windows service
Service logon account	Decide early whether the Agent service runs as Local System or a domain service account. A domain account is usually needed for network shares and for running jobs as other users
Run-as users	Jobs that run as a different Windows user need that user's credentials registered for the Agent. Use dedicated, non-expiring service accounts managed by your identity team
Paths	Avoid mapped drive letters in jobs. They are per-session and usually missing for a service. Use UNC paths
PowerShell	Call powershell.exe -NoProfile -ExecutionPolicy Bypass -File ... and make scripts return explicit exit codes
Antivirus / EDR	Exclude the Agent home directory from real-time scanning if your security team allows it. Scanning can slow job starts noticeably

Linux / UNIX Agent considerations

- Install as the dedicated Agent OS account (for example ctmagent), not as root, unless jobs must switch users. In that case follow BMC's root-mode instructions.
- Complete BMC's pre-installation procedures (limits, required packages) before installing.
- Make sure the Agent starts at boot, through the systemd unit or the startup script BMC provides.

- Keep enough free space in the Agent home: job output and logs accumulate there.

9. Managed File Transfer (MFT) administration

Control-M MFT runs file transfers as jobs, so transfers share dependencies, alerts and statistics with the rest of the workflow. It is a plug-in on the Agent.

Item	What the administrator sets up
MFT plug-in	Deployed on the Agents that will perform transfers
Connection profiles	One per remote endpoint: SFTP, FTP/FTPS, S3, Azure Blob, Google Cloud Storage, local, and others depending on version
Keys and certificates	SSH key pairs for SFTP, host key verification, and certificates for FTPS. Record expiry dates
PGP	Encryption/decryption settings when partners exchange encrypted files
Agentless hosts	Remote endpoints reached without installing an Agent there
Search and auditing	The Managed File Transfer domain shows every transfer: status, source, destination, run ID

MFT hygiene

- Name connection profiles after the partner and environment, for example SFTP_BANKX_PRD.
- Keep a register of partner keys and certificates with expiry dates. Expired keys are a common cause of night-time failures.
- Use host-key checking for SFTP. Turning it off hides man-in-the-middle risk.
- Prefer centralized connection profiles, so a password change is made once instead of on every Agent.

10. Migrating to Control-M SaaS

Most teams move from self-hosted Control-M to SaaS in stages instead of all at once.

Phase	Activities
Assess	Inventory folders, job types, plug-ins, Agents, calendars, users and integrations. Find unsupported or obsolete job types and scripts with hard-coded paths
Prepare	Set up the tenant, IdP, roles, Agent tags, and network access from Agent hosts to SaaS
Connect	Optionally use Unified View: connect existing self-hosted Control-M/Servers so operators can work from the SaaS interface early
Move Agents	Install SaaS Agents, or connect existing Agents to both self-hosted and SaaS servers during the transition
Move workload	Migrate definitions application by application (BMC migration tooling, Automation API export/import). Run parallel or shadow cycles for critical flows
Cut over	Hold the old definitions, enable the SaaS ones, and watch the first few New Day cycles closely
Decommission	Remove old Agents, servers and accounts. Archive old definitions

Other schedulers

Workloads coming from other schedulers (for example cron, Windows Task Scheduler, or other enterprise schedulers) can be converted with Control-M Conversion and the SaaS Self Conversion feature.

11. Security hardening checklist

Control	Recommended setting
Authentication	SAML 2.0 SSO with MFA enforced at the IdP. Internal users only for break-glass
Emergency access	Two to three named Emergency Users; test the Emergency URL every quarter
Least privilege	Separate roles for planning, L1, L2 and admin; application teams limited to their own folders
Sensitive actions	Set to OK, Kill, Skip, Bypass and Edit Script only for L2 and above; audit annotations required
Run-as users	Restrict which OS accounts each role can run jobs as; never allow interactive admin accounts
API tokens	One token per pipeline or integration, minimal roles, named after the owner, rotated on a schedule
Agent tokens	Generated only when needed; delete unused tokens
Connection profiles	Centralized, credentials rotated, separated per environment
Scripts	Stored on hosts with write access only for the owning team; no passwords in scripts or job command lines
Network	Only the required outbound access from Agent hosts; document proxy and TLS inspection exceptions

12. Backup, recovery and change control

BMC is responsible for backing up and recovering the SaaS platform. Your responsibility is the parts BMC cannot see: your Agents and hosts, and the ability to roll back your own definition changes.

- **Definition backups:** export folders regularly through the Automation API and commit them to Git. This gives you history and a fast way to restore a folder someone broke.
- **Agent resilience:** host groups plus documented Agent re-install steps (installer location, token tag, silent-install XML).
- **Configuration record:** keep a list of roles, Agent tags, connection profiles (without passwords), calendars, and notification destinations.

```
# Export folder definitions to JSON (example)
ctm deploy jobs::get -s "server=*&folder=FIN*" > backup/FIN_folders.json

# Export calendars
ctm deploy calendars::get -s "server=*&name=*" > backup/calendars.json
```

Change control for definitions

Change type	Suggested control
New flow or major redesign	Change request; tested in a non-production folder; peer review of JSON or workspace
Small change (time, host group, notification)	Standard change; check-in comment with the ticket number
Emergency fix to tonight's run	Edit in Monitoring (affects this run only), then raise a change to fix the definition
Calendar updates (e.g. holidays)	Annual task, done well before year end; second person checks the dates

13. Administrator troubleshooting

Symptom	Likely cause	What to check / do
Agent install says the token is invalid	Token deleted, mistyped, or for the wrong tag	Copy the token again from Agent Tokens; check the tag pattern matches
Agent installed but not listed	Outbound connection blocked	Proxy, firewall and DNS from the host; installer log in BMCINSTALL\log
Agent listed as Unavailable	Service stopped, host down, network change	Start the service, ping from the UI, then check connectivity
Plug-in job fails 'profile not found'	Connection profile missing for that Agent or environment	Use a centralized profile or define it for the Agent; check the name in the job
User can log in but sees nothing	Role has no folder or server authorizations	Check role membership and IdP group mapping
SSO login loop or error	Certificate or entity ID mismatch at the IdP	Compare IdP settings with the SAML values in Configuration; use the Emergency URL if needed
API call returns 401	Token deleted, expired, or wrong header	Send the token in the x-api-key header; check the token still exists
API call returns 403	Token's roles don't allow the action	Add only the needed authorization to the token's role
Windows job: access denied on a share	Agent service or run-as user lacks rights	Use UNC paths; check the account's share and NTFS permissions

Calling the REST API directly

Every SaaS Automation API call carries the token in the x-api-key HTTPS header.

```
# PowerShell: list Control-M/Servers
$endpoint = 'https://<your-endpoint>/automation-api'
$headers = @{ 'x-api-key' = $env:CTM_API_TOKEN }
Invoke-RestMethod -Uri "$endpoint/config/servers" -Headers $headers

# PowerShell: list Agents on a server
Invoke-RestMethod -Uri "$endpoint/config/server/<server>/agents" -Headers $headers
```

14. Onboarding a new application

A repeatable checklist keeps every new application consistent with the rest of the estate.

Step	Administrator task	Done with
1	Agree application code (e.g. FIN), owners, support group and escalation contacts	Application owner
2	Create or reuse Agent tag (e.g. FIN-PRD, FIN-NPD) and generate Agent tokens	Admin
3	Install at least two Agents per environment; create host groups (HG_FIN_PRD)	Infra team
4	Deploy required plug-ins; create centralized connection profiles per environment	Admin + app team
5	Register run-as service accounts and restrict them to the application's roles	Identity team
6	Create roles: App-Support-FIN (view, limited actions), FIN scheduler rights; map IdP groups	Admin
7	Create notification destinations (FIN-SUPPORT mail list) and viewpoint for operators	Admin
8	Agree calendars (working days, holidays) and the New Day impact on overnight flows	Scheduler
9	Build flows in non-production; test failure paths, reruns and notifications	Scheduler + app team
10	Production check-in with change ticket; brief L1/L2 with runbooks; watch the first cycles	All

15. Agent upgrade runbook

1. Read the release notes for the target Agent version and any plug-in compatibility notes.
2. Pick a quiet window. List the jobs that normally run on the Agent during that window.
3. If the Agent is in a host group, confirm the other members are Available, so jobs can route to them.
4. If the Agent is alone, hold its jobs for the window (or use a Workload Policy to pause them).
5. Stop the Agent first if MFT or Control-M for SAP is installed on it.
6. Upgrade interactively, silently with the XML file, or with the Automation API provision service (Agent 9.0.20.280 or later).
7. Check installed-versions.txt, start the Agent if needed, and Ping it from the Agents tab.
8. Run a test job (a Dummy or a simple command) on the Agent, then release held jobs.
9. Record the new version in your Agent inventory and close the change.

```
# Rough sequence with the Automation API (check exact syntax for your version)
ctm provision upgrades:versions::get          # available versions
ctm provision upgrade::install <server> <agent> Agent <version>
ctm provision upgrades::get                  # track progress
```

16. Glossary

Term	Meaning
Agent	Process on your host that runs jobs and returns results to SaaS
Agent tag	Label that groups Agents for authorization; one per Agent
Agent token	Secret used to install an Agent and connect it to your tenant; tied to a tag
API token	Secret used by scripts and pipelines to call the Automation API; carries roles
Centralized connection profile	Credentials and endpoint for a plug-in, defined once in SaaS
Emergency user	Admin user who can sign in through the Emergency URL when the IdP is down
Host group	Set of Agents a job can run on; gives load balancing and failover
Plug-in	Agent extension that adds job types for an application or service
Site Standard	Rule that enforces naming and property conventions on definitions
Tenant	Your organization's isolated Control-M SaaS environment
Unified View	Mode that connects self-hosted servers and z/OS to the SaaS interface

Based on BMC's Control-M SaaS documentation (documents.bmc.com/supportu/controlm-saas), checked October 2026. Control-M SaaS was previously called BMC Helix Control-M. Menu names and limits change between releases, so confirm details against your tenant and the current docs.